

Subhodip Panda

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Research Interests

My broader research interest lies in trustworthy/reliable machine learning, especially Differential Privacy, Machine Unlearning, Data Influence Estimation, Conformal Prediction, Statistical Learning Theory, and Information Theory.

Education

- 2022–Present **Ph.D in Engineering, Indian Institute of Science, Bangalore**, Representation Learning Lab, Department of ECE, Advisor: Dr. Prathosh A.P, GPA: 8.8/10.
Thesis: Privacy and Uncertainty Aware Learning Algorithms
- 2020–2021 **P.G. Diploma in Statistics, Indian Statistical Institute, Chennai**, Statistical Science Unit, Advisor: Dr. Sudheesh Kumar K., Grade: 73% (3rd in Class of 42).
Graduate (Master's) course works on Statistics and Mathematics
- 2013–2017 **B.Tech in ECE, Indian Institute of Engineering Science and Technology, Shibpur, Kolkata**, Advisor: Dr. Monojit Mitra, GPA: 8.54/10 (9th in Class of 48).
Undergraduate course works in electronics and communication engineering

Work & Internships Experience

- May,2026– Present (Internship) **Microsoft, APPLIED RESEARCH SCIENTIST INTERN, Bangalore, India.**
○ **Personalization of Time Scheduling through LLM agents:** The objective of this project is to develop an agentic workflow that enhances the existing Microsoft Time Scheduling API by enabling personalized calendar scheduling experiences. The work involves understanding and extending the current scheduling pipeline to incorporate user-specific preference profiles, while leveraging fine-tuned Large Language Models (LLMs) within the decision-making loop to facilitate intelligent, context-aware scheduling.
- May,2025– Aug,2025 (Internship) **Adobe Research, PH.D RESEARCH INTERN, Bangalore, India.**
○ **Data Attribution for VLMs:** The objective of this work is to attribute influence scores to training samples for an inference-time generated image from Vision Language Models (e.g., Stable Diffusion). The primary challenge of this work is to solve the scalability problems of data attribution techniques for large-scale models and datasets.
- Aug,2021– Dec,2021 (Full time) **Oneirix Labs, RESEARCH ASSOCIATE, Pune, India.**
○ **3D Modeling for Medical Imaging:** 1. Developed several algorithms that are used for 3D modeling of the human heart using Python. 2. Developed an algorithm to detect heart valves from a sequence of ultrasonic images/videos. 3. Created one algorithm that will sample from a small no of ultrasonic heart images to make a longer sequence of images with a smooth transition.
○ **Simulation and Modeling:** Implemented the task of determining the kinetics constants of the adsorption reaction in the Langmuir Adsorption Model. This model aims to fit a non-linear curve using Levenberg-Marquardt (LM) Algorithm.
- Aug.2017– Aug.2019 (Full time) **Larsen and Toubro, SENIOR ENGINEER, Chennai, India.**
○ **Nagpur and Jaipur Smart City Project:** Advisory role for the local Government developing action plans that would lead forward a path in curbing pollution and traffic. Time series analysis and data visualization of environmental sensor's data that was captured over one year across different locations of the city.

May,2016– **Indian Space Research Organization (ISRO)**, RESEARCH INTERN, *Bangalore, India*.
 Aug,2016 (Internship) ○ **Project: Driving a Charged Coupled Device by the microcontroller for LIBS based applications**
 CCD (Charged Coupled Device) which had been used for LIBS (Laser Induced Breakdown Spectroscopy) based applications was analyzed in this project. LIBS had been used to generate a high-powered laser beam that was focused on a soil sample and the output was captured by CCD. The CCD was driven by the microcontroller and using a Bluetooth module the soil sample data was stored in a Bluetooth-enabled Android device which was later analyzed for chemical molecule detection. This project has applications for agricultural purposes. ■ **P.1 is the published paper of this project.**

Publications

P=Published, S=Submitted

- P.9 **Regret Tail Characterization of Optimal Bandit Algorithms with Generic Rewards**, *Subhodip Panda, Shubhada Agrawal*, IEEE INTERNATIONAL SYMPOSIUM ON INFORMATION THEORY, 2026, [Paper link](#).
Description: This work establishes novel regret tail bounds for asymptotically optimal multi-armed bandit algorithms under general reward distribution classes, providing a theoretical characterization of their tail behavior beyond expected regret.
- P.8 **f-INE: A Hypothesis Testing framework for Estimating Influence under Training Randomness**, *Subhodip Panda, Dhruv Tarsadiya, Shashwat Sourav, Prathosh A.P., Sai Praneeth Karimireddy*, THE 14th INTERNATIONAL CONFERENCE ON LEARNING REPRESENTATIONS, 2026, [Paper link](#).
Description: We develop a novel hypothesis testing framework for data attribution, enabling the identification of influential training samples underlying model predictions and generative outputs. We validate the proposed approach through comprehensive experiments on vision classification benchmarks and instruction-tuned Large Language Models (LLMs), demonstrating its effectiveness and scalability.
- P.7 **Unlearning in Diffusion Models under Data Constraints: A Variational Inference Approach**, *Subhodip Panda, MS Varun, Shreyans Jain, Sarthak Maharana, Prathosh A.P.*, TRANSACTIONS ON MACHINE LEARNING RESEARCH (TMLR), 2026, [Paper link](#).
Description: We develop a theoretically motivated variational inference framework for concept unlearning in diffusion models, enabling the selective removal of both class-level concepts and fine-grained semantic features from Denoising Diffusion Probabilistic Models (DDPMs) and Stable Diffusion models.
- P.6 **Partially Blinded Unlearning: Class Unlearning for Deep Networks a Bayesian Perspective**, *Subhodip Panda, Shashwat Sourav, Prathosh A.P.*, THE 39th ANNUAL AAAI CONFERENCE ON ARTIFICIAL INTELLIGENCE, 2025, [Paper link](#).
Description: This work develops a novel Bayesian framework for machine unlearning, providing a principled probabilistic formulation of the forgetting process. Building on this framework, we propose a scalable algorithm for efficient class unlearning in deep vision classifiers while preserving performance on retained classes.
- P.5 **Concept Forgetting via Label Annealing**, *Subhodip Panda, Ananda Theertha Suresh, Atri Guha, Prathosh A.P.*, THE 41st CONFERENCE ON UNCERTAINTY IN ARTIFICIAL INTELLIGENCE (UAI), 2025, [Paper link](#).
Description: This work overcomes a fundamental limitation of existing machine unlearning approaches by enabling the removal of distributed semantic concepts rather than isolated training samples. Specifically, we develop a scalable LAN-based algorithm for concept forgetting, capable of forgetting concepts that are implicitly represented across all training examples.
- P.4 **Adapt Then Unlearn: Exploiting Parameter Space Semantics For Unlearning In Generative Adversarial Networks**, *Piyush Tiwari, Atri Guha, Subhodip Panda, Prathosh A.P.*, TRANSACTIONS ON MACHINE LEARNING RESEARCH (TMLR), 2025, [Paper link](#).
Description: This work introduces one of the first approaches to machine unlearning in pre-trained Generative Adversarial Networks (GANs), enabling the selective removal of subtle semantic image features while preserving the quality of generated images.

- P.3 **FAST: Feature Aware Similarity Thresholding for weak unlearning in black-box generative models**, *Subhodip Panda and Prathosh A.P.*, IEEE TRANSACTION ON ARTIFICIAL INTELLIGENCE (TAI), 2024, [Paper link](#).
Description: This work addresses a fundamental challenge in machine unlearning: how to selectively forget concepts from a black-box generative model without access to its parameters. Motivated by practical deployment scenarios where model parameters are inaccessible due to proprietary or intellectual property constraints, we propose a feature-aware blocking algorithm that enables weak unlearning in black-box generative models.
- P.2 **Variational Diffusion Unlearning: a variational inference framework for unlearning in diffusion Models**, *Subhodip Panda, MS Varun, Shreyans Jain, Sarthak Maharana, Prathosh A.P.*, NEURIPS SAFE GENERATIVE AI WORKSHOP, 2024, [Paper link](#).
- P.1 **Driving a Charged Coupled Device (CCD) by micro-controller for LIBS based application**, *Avijit Mandal, Subhodip Panda, Adwaita Goswami*, IEEE INTERNATIONAL SYMPOSIUM ON DEVICES, CIRCUITS, AND SYSTEMS (ISDCS), 2018, [Paper link](#).

Projects and Certifications

- Ph.D Course Project **Online Changepoint Detection Algorithms:** Change Point Detection refers to the task of identifying abrupt changes or shifts in time series data. This project investigates the sequential hypothesis testing framework to analyze traditional CUSUM statistic-based change point detection algorithms. Furthermore, it conducts a comparative study of neural change point detection methods to evaluate their effectiveness and performance relative to classical approaches.
- Master's Project **Automatic categorization and segmentation of land cover using satellite images for Sustainable Urban Development:** Automatic categorization and segmentation of land cover is of great importance for sustainable development, autonomous agriculture, and urban planning. In this project, different image segmentation-based architectures have been explored and used for land cover image segmentation. Specifically, DeepLab-V3 is used as the final model for the segmentation task.
- Online Certifications Coursera DeepLearning.AI specialization: Neural Networks, CNN, Sequence Models, Coursera Mathematics for Machine Learning specialization, Coursera SQL for Data Science

Skills

- Programming Python, R, SQL
- ML Libraries Pytorch(Preferred), Tensorflow, Keras, Scikit-learn
- Scripting L^AT_EX, MS Word, MS PowerPoint
- Other Tools Github/Gitlab, Arduino, Atmel-ATMEGA, Linux

Awards and Achievements

- Fellowships and Grants
- Recipient of prestigious Google and KIAC research travel grant for attending ICLR, 2026.
 - Recipient of Kotak-IISc AI Center research travel grant for attending AAAI, 2025.
 - [Recipient of VeARC PhD Fellowship](#) for research excellence (Top 3 at IISc) in 2024-25.
 - Recipient of Ministry of Education (MoE) Fellowship as a part of doctoral research.
 - Recipient of GABESU Scholarship 2015 for academic excellence at IEST, Shibpur.
- Other Recognitions
- Winner of best poster presentation in the **ACM ARCS 2026** event held at IIT, Hyderabad.
 - Secured AIR-10 in the Indian Statistical Institute's PGDSMA entrance examination.
 - Scored 99.5+ percentile (State Rank: 432) in JEE MAINS, 2013 examination.
 - Ranked top 0.6 percent (Rank: 830) in State Engineering entrance exam (WBJEE), 2013.

Academic Works and Responsibilities

- Reviewer NeurIPS(2023), ICLR(2024), ICML(2024), AAAI(2025)

Teaching Teaching Assistant for Graduate Courses: 1. Pattern Recognition and Neural Network (E2-233)
2. Advanced Deep Representation Learning (E2-333)

Graduate Courses Real Analysis, Linear Algebra, Multi-variable Calculus, Probability Theory, Detection and Estimation theory, Advanced Image Processing, Pattern Recognition and Neural Networks, Advanced Studied Deep Representation Learning, Online Prediction and Learning, Concentration Inequality, Measure theory

Student Mentoring

2023-2024 Atri Guha (B.Tech @ IIT, Patna) → (ML Research Engineer @ Tiger Analytics)

2024-2025 MS Varun (B.Tech @ PES University) → (ML Engineer @ VISA)

2024-Present Shahswat Sourav (BS @ IISER, Bhopal) → (Ph.D @ University of Washington)